1 Calculate Joint Probabilities in HMM

$\theta = (\pi, A, E)$

$P(x,z|\theta) = p(z_1)\,p(x_1|z_1)\,(p\, z_2|z_1)\,p(x_2|z_2) \cdots p(z_n|z_{n-1})\,p(x_n|z_n)$

$$\begin{array}{ccccccccccc} & x_1 & & x_2 & & x_3 & & x_4 & & x_5 \\ x = & 6, & & 3, & & 1, & & 2, & & 4 \end{array}$$

$$\begin{array}{cccccc} z = & \text{Load}, & \text{Fair}, & \text{Fair}, & \text{Load}, & \text{Fair} \\ & z_1 & z_2 & z_3 & z_4 & z_5 \end{array}$$

Time t	Observation x_t (Roll)	Hidden State z_t (Die Used
1	6	Load
2	3	Fair
3	1	Fair
4	2	Load
5	4	Load

$p(x,z) = P(L) \times P(6|L) \times P(F|L) \times P(3|F) \times P(F|F) \times P(1|F) \times P(L|F) \times P(2|L) \times P(L|L) \times P(4|L)$

$p(x,z) = 1/2 \times 1/2 \times 0.4 \times 1/6 \times 0.6 \times 1/6 \times 0.4 \times 1/10 \times 0.6 \times 1/10$

$p(x,z) = 0.000004$

$= 4 \times 10^{-6}$

∴ The probability of this scenario

x = 6, 3, 1, 2, 4

z = Load, Fair, Fair, Load, Fair

is $p(x,z) = 4 \times 10^{-6}$

2 Viterbi Algorithm

The V Matrix

	t = 1	t = 2	t = 3
k = 1 (Fair)	$\frac{1}{6} * \frac{1}{2}$	$1/6 \times 1/20 = 1/120$	$1/6 \times 1/120 = 1/900$
k = 2 (Loaded)	$\frac{1}{10} * \frac{1}{2}$	$1/2 \times 1/30 = 1/60$	$1/10 \times 1/100 = 1/1000$
observation	"4"	"6"	"3"

The Ptr Matrix

	t = 1	t = 2	t = 3
k = 1 (Fair)	0	1	2
k = 2 (Loaded)	0	1	2
observation	"4"	"6"	"3"

Traceback

t = 3 → Fair

t = 2 → Loaded

t = 1 → Fair

When t = 2 and k = 1 (Fair State) $x = 6$

$V_{fair}(1) \times P(F|F) = 1/12 \times 0.6 = 1/20$ ← max => $1/6 \times 1/20 = 1/120$

$V_{loaded}(1) \times P(F|L) = 1/20 \times 0.4 = 1/50$

When t = 2 and k = 2 (Loaded State)

$V_{fair}(1) \times P(L|F) = 1/12 \times 0.4 = 1/30$ ← max => $1/2 \times 1/30 = 1/60$

$V_{loaded}(1) \times P(L|L) = 1/20 \times 0.6 = 3/100$

When t = 3 and k = 1 (Fair State) $x = 3$

$V_{fair}(2) \times P(F|F) = 1/120 \times 0.6 = 1/200$

$V_{loaded}(2) \times P(F|L) = 1/60 \times 0.4 = 1/150$ ← max => $1/6 \times 1/150 = 1/900$

When t = 3 and k = 2 (Loaded State)

$V_{fair}(2) \times P(L|F) = 1/120 \times 0.4 = 1/300$

$V_{loaded}(2) \times P(L|L) = 1/60 \times 0.6 = 1/100$ ← max => $1/10 \times 1/100 = 1/1000$

$p(x, z^*) = \max(t=3) = 1/900$

∴ The $p(x, z^*)$ is exactly $1/900$ (the largest value from the V matrix) and the hidden state sequence z^* found is : Fair → Loaded → Fair